

LUIS MIGUEL HERRERA

COMPUTATIONAL SOCIAL DATA SCIENTIST

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SUMMARY

Graduate in Computational Social Science and Political Science, specialized in data modeling, advanced analytics, and predictive tools to decode complex organizational and socioeconomic dynamics. Extensive experience in developing ETL processes and using Python and SQL to transform large volumes of unstructured data into clear and actionable solutions. Focused on applying statistical rigor to eliminate bias and generate evidence-based insights, with the goal of accelerating internal innovation and optimizing strategic decision-making.

PROJECTS

Predictive Modelling of Multidimensional Poverty in Argentina (2016–2025)

- Developed a high-performance predictive framework to classify multidimensional poverty using EPH survey data, using XGBoost, Random Forest, and CART models.
- Implemented an end-to-end Tidymodels pipeline, including SMOTENC for class imbalance (handling both categorical and numerical features) and MCA (Multiple Correspondence Analysis) for advanced feature engineering.
- Eliminated structural data leakage by deconstructing the poverty index (MPI) and identifying circular predictors, ensuring the model learned real socio-economic patterns rather than arithmetic identities.
- Leveraged Explainable AI (XAI) tools, including SHAP, LIME, and Breakdown plots, to transform "black-box" predictions into actionable insights regarding socio-economic drivers.
- Optimized model performance using Racing Methods (ANOVA) for hyperparameter tuning, significantly reducing computational overhead while maintaining statistical robustness via K-fold cross-validation.

Real-Time Grocery Index – Team Project

- Built a "Live CPI" by developing a scraper in R to extract real-time prices for essential goods from major supermarkets (e.g., Mercadona, Carrefour).
- Normalized heterogeneous data (e.g., liters vs. pints) using R code and specialized packages such as rvest, httr, and jsonlite.
- Created a standardized "Basket Total" dashboard to track short-term inflation trends, bypassing the time lags of official statistics.

Internet Use and COVID-19 Attitudes in Europe

- Conducted a multifocal analysis using the European Social Survey (ESS) Round 10 for 22 countries, running multiple logistic regressions.
- Tested the causal relationship between internet usage and political internal efficacy, government support, and denialist behaviors during the pandemic.
- Handled complex nested data structures and balanced structural imbalances to mitigate social-desirability bias in public opinion data.

EDUCATION

University Carlos III de Madrid	2025 – 2026
Master's in Computational Social Sciences	
University Carlos III de Madrid	2020 – 2025
Bachelor's Degree in Political Science and International Studies	
University of Buenos Aires	March 2024 – June 2024
Exchange Program in Political Science	

LANGUAGES

 Spanish (Native)	 English (IELTS – C1)	 French (Basic – Professional Working)
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TECHNICAL SKILLS

Programming & Data Management

- Python (Pandas, Scikit-Learn, NumPy)
- R (Tidyverse, Tidymodels, rvest)
- SQL (Data querying and management)

Data Science & Machine Learning

- Machine Learning Models
- Explainable AI (XAI)
- Feature Engineering
- Cross-validation & Hyperparameter Optimization

Development & Tools

- Visual Studio Code (VS Code)
- AI Agent Management (Cline)
- Git & GitHub

Data Visualization & Analytics

- Shiny
- Tableau
- Power BI (In Progress)

ONGOING EDUCATION & CERTIFICATIONS

- AWS Cloud Practitioner | EC2 Essentials (In Progress)
- Data Visualization with Power BI (In Progress)
- Agile Explorer by IBM
- AI Fundamentals by IBM